

Morphology-Preserving Virtual Staining from H&E to PAS and IHC using Enhanced Diffusion Models

Giuseppe Spathis¹ , Imen Chtourou¹ , Xavier Descombes¹ , Damien Ambrosetti² , Giorgio Toni² , and Paul Hannebel² 

¹ Université Côte d’Azur, INRIA, CNRS, Sophia Antipolis, France

² Department of Pathology, CHU Nice, Université Côte d’Azur, Nice, France
giuseppe.spathis@inria.fr

Abstract. Generating specialized histopathological stains like Immunohistochemistry (IHC) and Periodic Acid-Schiff (PAS) in a laboratory is costly and time-consuming, making purely digital virtual staining a critical solution. However, acquiring perfectly paired training data is physically impossible, and standard diffusion models often alter the underlying tissue architecture when applied to challenging unpaired datasets. To address this, a novel morphology-preserving virtual staining framework is proposed for unpaired H&E-to-PAS and H&E-to-IHC translation. For the PAS domain, a luminance-guided post-processing step is introduced to recover fine structural boundaries. For the IHC domain, a two-step inference pipeline guides the generative process using a nuclear mask to protect spatial fidelity, followed by an automated refinement step to remove structural artifacts. Quantitative and qualitative evaluations demonstrate that the proposed approach improves morphological metrics compared to baselines. Furthermore, incorporating these generated virtual stains into a downstream subcancer classification task improves overall accuracy, demonstrating the practical and clinical utility of the proposed method.

Keywords: Virtual Staining · Computational Pathology · Generative AI · Unpaired Image Translation · Morphology Preservation.

1 Introduction

In computational pathology, Hematoxylin and Eosin (H&E) is the standard staining protocol for tissue examination. However, specialized stains such as Periodic Acid-Schiff (PAS) or Immunohistochemistry (IHC) are frequently required to highlight specific cellular structures and biomarkers that remain invisible in standard H&E slides [3, 9]. Generating these specialized stains in a wet lab is a costly, time-consuming process, and many clinical centers lack the facilities to perform them routinely [1, 2]. Consequently, virtual staining—the purely digital transformation of H&E images into specialized stains without the need for chemical reagents—has emerged as a critical solution to accelerate and democratize the histopathological workflow.

Historically dominated by Generative Adversarial Networks, virtual staining is increasingly adopting Diffusion Models due to their superior visual fidelity and morphology preservation, despite higher computational costs [8, 11].

A fundamental challenge in virtual staining is the impossibility of obtaining perfectly paired data, as different chemical stains cannot be applied to the exact same tissue slice. Consequently, existing methods rely on either weakly paired data from consecutive cuts—which inherently suffer from physical deformations—or completely unpaired datasets. This work focuses on the highly challenging *unpaired* setting. Because standard diffusion models often alter tissue architecture in the absence of paired data, a novel morphology-preserving framework is introduced for unpaired H&E-to-PAS and H&E-to-IHC translation. In this paper, we present a unified framework addressing tasks in multicentric pathology data. Our key contributions include: (i) a complete pipeline for H&E-to-PAS virtual staining using the His-MMDM diffusion model, incorporating linear normalization, gamma correction, and a novel luminance-guided structural correction; and (ii) a two-step morphology-preserving inference strategy for unpaired H&E-to-IHC translation based on the PixCell-1024 foundation model, which integrates a lock-release nuclear mask mechanism and post-hoc automated removal of hallucinated artifacts.

2 Methodology

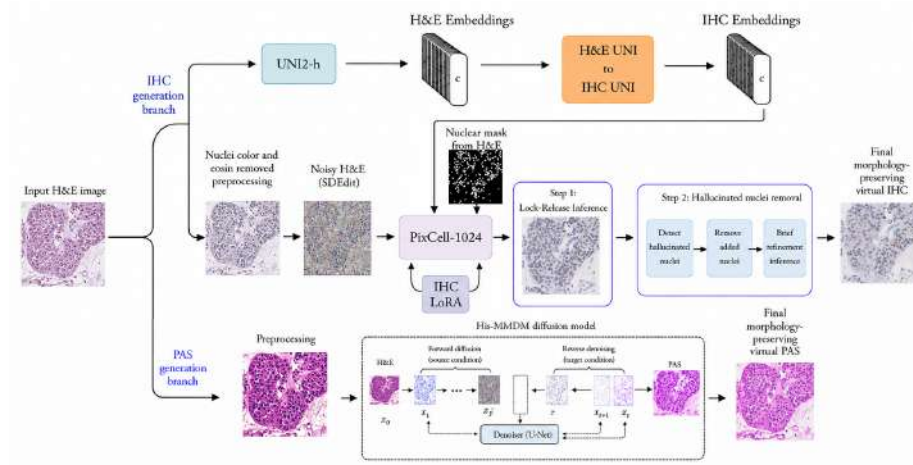


Fig. 1. Overview of the proposed morphology-preserving virtual staining pipeline.

Figure 1 illustrates the overall architecture of the proposed virtual staining framework. The pipeline is divided into two specialized branches H&E to PAS and H&E to IHC each tailored to address the unique colorimetric and structural

challenges of the respective target domain while strictly preserving the original tissue morphology.

2.1 Virtual Staining from H&E to PAS

Preprocessing. Raw whole-slide image patches from both HE and PAS modalities undergo a two-step normalization procedure prior to diffusion model training. Let $\{x_i\}_{i=1}^N$ denote the pixel intensities of a patch, and let Ω denote the set of tissue pixels (background excluded via Otsu thresholding). We compute the intensity range as:

$$\text{PAS}_{\text{Low}} = \min\{x_i : i \in \Omega\}, \quad \text{PAS}_{\text{High}} = \max\{x_i : i \in \Omega\}. \quad (1)$$

Linear stretching followed by gamma correction is applied to each pixel $x \in [0, 255]$:

$$x' = 255 \cdot \left[\text{clip} \left(\frac{x - \text{PAS}_{\text{Low}}}{\text{PAS}_{\text{High}} - \text{PAS}_{\text{Low}}}, 0, 1 \right) \right]^\gamma, \quad (2)$$

where $\text{clip}(\cdot, 0, 1)$ saturates the value with $[0, 1]$ and γ is a stain-specific contrast correction exponent tuned in a held-out validation set. This normalization reduces inter-scanner and inter-center variability, a critical requirement for multicentric datasets.

Diffusion Model. We adopt His-MMDM [8] as our base generative model, which is trained in an unpaired fashion using a multi-modal diffusion objective. His-MMDM does not require pixel-aligned pairs of HE and PAS patches, making it directly applicable to our clinical dataset where serial sections are not systematically acquired.

Post-Processing for Structural Preservation. A key limitation of raw diffusion-generated PAS images is the introduction of structural artifacts at tissue boundaries – fine tubular and glomerular outlines are sometimes displaced or blurred compared to the original HE image. To correct this, we introduce a luminance-guided post-processing step operating in the LAB color space as follows: **(i) Edge extraction:** Canny edge detection is applied to the grayscale HE image to produce a binary edge map E_{HE} . **(ii) Guided filter:** A guided filter [6] is applied to the luminance channel L of the generated PAS image, using the grayscale HE image as the guidance signal. This corrects the generated luminance while preserving the PAS chromatic channels a and b unchanged. **(iii) Local correction:** The final luminance is computed as a locally-weighted blend:

$$\text{band}(x) = \mathcal{G}_{\sigma_b}(\delta(E_{\text{HE}}))(x)$$

$$L_{\text{final}}(x) = (1 - \text{band}(x)) \cdot L_{\text{original}}(x) + \text{band}(x) \cdot L_{\text{guided}}(x) \quad (3)$$

where E_{HE} denotes the binary Canny edge map extracted from the grayscale H&E image, $\delta(\cdot)$ denotes morphological dilation using an elliptical structuring

element, and $\mathcal{G}_{\sigma_b}$ denotes Gaussian smoothing with standard deviation σ_b . The resulting $\text{band}(x) \in [0, 1]$ is a soft edge-weighting function that assigns higher weights to pixels located near tissue boundaries and gradually decreases toward zero away from the edges. Consequently, the guided-filter correction is applied primarily around structural contours while preserving the original diffusion-generated luminance elsewhere.

This strategy selectively corrects the generated image only at tissue boundaries, where structural fidelity is most diagnostically critical, leaving the global color appearance unchanged.

2.2 Virtual Staining from H&E to IHC

The proposed approach is based on PixCell [16], a Latent Diffusion Model (LDM) conditioned on the embedding of the input image. As illustrated in the IHC branch of Figure 1, a tailored two-step inference pipeline is introduced to preserve the morphology of the H&E image.

To initialize the process, pure-noise sampling is replaced by SDEdit [10], injecting controlled noise directly into the input H&E image. A preprocessing step is then applied to prevent unnatural color shifts in the stroma and correctly guide the nuclei towards the target IHC blue: the eosin channel is digitally stripped and the nuclei undergo Reinhard color normalization [12] using an IHC reference.

First Inference Step. Spatial fidelity is enforced through a StarDist-guided *lock-and-release* mechanism [14]. Let $M \in [0, 1]$ be the soft nuclear mask, $\hat{x}_{t-1}$ the predicted denoised latent, and $\odot$ denote the element-wise Hadamard product. For the initial n steps of the reverse diffusion process, the updated latent x_{t-1} is spatially locked to the noise-injected H&E latent x_{t-1}^{HE} :

$$x_{t-1} = M \odot x_{t-1}^{\text{HE}} + (1 - M) \odot \hat{x}_{t-1} \quad (4)$$

After n steps, the nuclei are “released,” allowing unconstrained denoising. This affords the LDM flexibility for color harmonization while strictly preserving morphology. To mitigate DAB variability, five candidates are generated. The final image is chosen by selecting the one that best matches a majority-vote DAB mask (where at least 3 out of 5 seeds agree on the stain presence), while penalizing artifacts in non-tissue regions.

Second Inference Step. This automated refinement pass removes structural artifacts. Occasional background hallucinations (false nuclei) are explicitly detected by reapplying StarDist to the generated image and performing a logical subtraction against the original H&E nuclear mask. The identified false nuclei are digitally erased using a baseline stromal color, smoothed via a 9×9 Gaussian kernel, and seamlessly harmonized through a highly localized SDEdit inference pass. By locking the validated regions of the image, the repaired patches are flawlessly integrated, yielding a morphologically consistent IHC translation.

3 Experiments and Results

3.1 Datasets and Implementation Details

The experiments utilized a multicentric private dataset of kidney cancer across four distinct renal tumor subtypes, comprising standard H&E images from CHU Nice and IHC CK7 images from a hospital in Lyon. Crucially, the patient cohorts from the two institutions are completely distinct, ensuring a strictly unpaired setting. Data were partitioned at the patient-level (4 train, 1 test per subtype), yielding 1,200 patches per modality for each subtype (4,800 H&E and 4,800 IHC patches in total) after filtering uninformative regions and artifacts.

Given the data scarcity in specialized medical cohorts, the pre-trained PixCell foundation model [16] was fine-tuned for unpaired H&E to IHC CK7 translation, serving as the baseline for all subsequent morphological preservation analyzes. To prevent catastrophic forgetting and ensure computational efficiency, the domain adaptation strategy trained two lightweight components while keeping the PixCell backbone strictly frozen: a Low-Rank Adaptation (LoRA) module—which injects a small fraction of trainable parameters into the attention layers to efficiently capture CK7-specific visual statistics without altering the foundation weights—trained on target IHC patches for 20 epochs, and a Multilayer Perceptron mapping network trained on both domains for 100 epochs. The H&E-to-PAS experiments were conducted on a multicentric dataset consisting of fifteen whole-slide images (WSIs), derived from five renal biopsies, with three serial sections available for each biopsy. The dataset was partitioned at the biopsy level to avoid data leakage, with four biopsies (twelve WSIs) used for training and one biopsy (three WSIs) reserved for testing. From these slides, we extracted 5,000 unpaired H&E patches and 5,000 unpaired PAS patches. The His-MMDM model was fine-tuned for this domain using these patches. Private datasets from the two institutions were de-identified and used in accordance with local institutional approval procedures. All experiments for both staining branches were executed on an NVIDIA A100-based high-performance computing cluster.

3.2 Evaluation Metrics

We assess perceptual quality using FID [7] and KID [4], measuring feature-space similarity between the generated stains and the real target images. To evaluate structural preservation, we compare the generated images directly with their paired H&E inputs. For color and texture shifts, we compute the 1D Wasserstein Distance (Earth Mover’s Distance) [13] between the flattened 256-bin density histograms averaged across the three color channels, Edge IoU (spatial overlap of Canny edge maps), and Nuclear IoU (spatial overlap of nuclear instance segmentation masks extracted via InstanSeg [5]).

The metric selection was adapted to the biological constraints of each domain. Edge IoU was evaluated exclusively for the PAS domain, as it prioritizes the sharpness of tubular and glomerular basement membranes. Conversely, Nuclear IoU was restricted to IHC: since CK7 is a membranous and cytoplasmic

Table 1. Quantitative results for HE-to-PAS virtual staining. ↓: lower is better; ↑: higher is better.

Method	FID ↓	KID ↓	Wasserstein ↓	Edge IoU ↑
Xiong et al. [15]	156.037	0.15294	10.926	0.571
Ours (w/o post-proc.)	87.060	0.09248	8.581	0.433
Ours (w/ post-proc.)	122.854	0.119	6.798	0.700

marker, strict nuclear preservation is diagnostically critical to accurately contextualize the DAB signal. Wasserstein distance was omitted for DAB-positive IHC subtypes to avoid signal interference. Finally, we tailored the downstream validation to the clinical role of each stain: we conducted a blind test for PAS to validate the visual authenticity and clinical plausibility of the generated stains, while for IHC, we performed downstream subcancer classification to demonstrate the preliminary evidence of utility of the generated biomarkers.

3.3 H&E to PAS Virtual Staining with Structural Post-Processing

Quantitative Comparison. Table 1 reports the quantitative results. We reimplemented the dual-path prompted inversion method of Xiong et al. [15] on our dataset. Our method without post-processing already outperforms this baseline on all distributional metrics. After post-processing, the Mean Wasserstein Distance improves further to 6.798 and the Mean Edge IoU reaches 0.700 – a 22% relative improvement over the baseline – demonstrating that our structural correction step substantially enhances the morphological quality of generated PAS images. As expected, the guided filter correction slightly reduces global distributional scores (FID and KID) due to the luminance modification, reflecting the inherent trade-off between perceptual realism and structural fidelity.

Qualitative Results. Visually, generated PAS images faithfully reproduce the characteristic purple-pink hue of the PAS stain and successfully reveal basement membrane outlines. After post-processing, tissue boundaries are sharper and better aligned with the original HE anatomy, while color consistency across the patch is maintained.

To further assess the visual quality of the generated PAS images, we conducted a blind test challenge in which generated and real PAS patches were mixed and presented to expert pathologists without labels. Figure 2 shows a representative set of eight patches drawn from this blind test, spanning a variety of renal tissue morphologies including tubular, glomerular and interstitial structures. The visual plausibility of these images across diverse tissue contexts demonstrates that the generated PAS staining is sufficiently realistic to be evaluated alongside authentic histological slides.

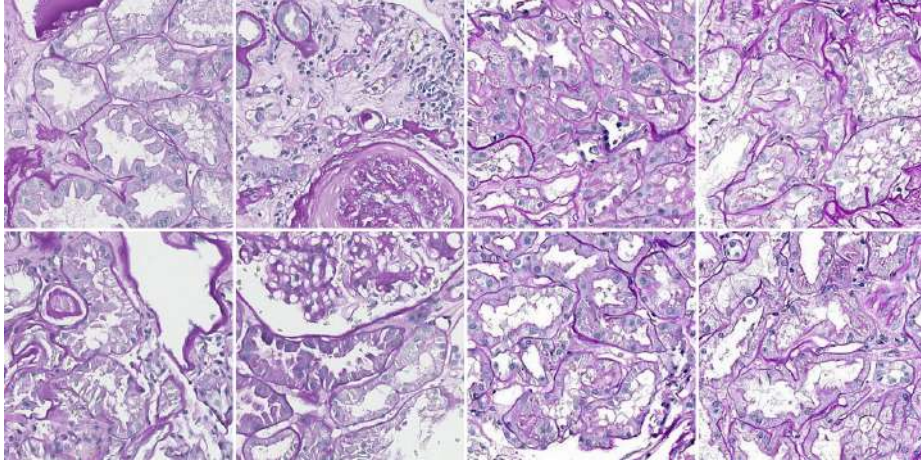


Fig. 2. Blind test challenge: a representative selection of eight PAS patches (real and virtually generated, unlabelled) presented to pathologists for evaluation. Images span diverse renal tissue morphologies. The generated patches show visual plausibility and staining consistency across diverse renal tissue morphologies.

3.4 Morphology-Preserving IHC Translation via Two-Step Inference

The proposed two-step inference pipeline effectively enhances spatial accuracy and structural preservation, as evidenced by Table 2. Globally, the approach prioritizes structural fidelity, substantially improving the Nuclear IoU metric for the evaluated subtypes, while yielding a slightly higher global FID score.

As illustrated in Figure 3, the proposed method qualitatively mitigates structural hallucinations and preserves biological fidelity. It correctly maintains a DAB-negative profile for Clear Cell and Oncocytoma, while precisely localizing CK7 expression—membranous in Chromophobe and cytoplasmic in Papillary—in DAB-positive tissues. However, it exhibits a slight colorimetric shift in the stromal background, which appears whiter compared to the grayish hue of real IHC slides. This indicates a limitation in fully harmonizing the global background noise distribution; nevertheless, as demonstrated in the subsequent downstream classification task, this visual discrepancy does not negatively impact diagnostic utility.

3.5 Downstream Application: Subcancer Classification

The practical utility of virtual stains was evaluated on a downstream subcancer classification task using an ImageNet pre-trained VGG16 model. To ensure robustness, the network was fine-tuned and assessed via a 5-fold cross-validation strategy with strict patient-level data partitioning.

Table 2. Quantitative comparison (mean $\pm$ std). Global represents the arithmetic mean across all tumor subtypes. Best results are in **bold**.

Subtype	Method	FID ($\downarrow$)	KID ($\downarrow$)	Wass. ($\downarrow$)	Nuc. IoU ($\uparrow$)
Clear Cell	Baseline	101.84 $\pm$ 1.26	0.114 $\pm$ 0.002	53.01 $\pm$ 0.05	0.09 $\pm$ 0.001
	Proposed	100.26 $\pm$ 0.55	0.110 $\pm$ 0.002	19.55 $\pm$ 0.11	0.25 $\pm$ 0.004
Chromo	Baseline	146.62 $\pm$ 1.52	0.145 $\pm$ 0.003	N/A	0.04 $\pm$ 0.001
	Proposed	158.42 $\pm$ 1.52	0.155 $\pm$ 0.002	N/A	0.07 $\pm$ 0.003
Onco	Baseline	162.67 $\pm$ 2.15	0.199 $\pm$ 0.003	30.81 $\pm$ 0.08	0.08 $\pm$ 0.001
	Proposed	139.23 $\pm$ 0.94	0.159 $\pm$ 0.002	20.12 $\pm$ 0.06	0.24 $\pm$ 0.002
Papillary	Baseline	111.98 $\pm$ 0.65	0.096 $\pm$ 0.002	N/A	N/A
	Proposed	131.99 $\pm$ 0.79	0.128 $\pm$ 0.003	N/A	N/A
Global	Baseline	130.78 $\pm$ 25.47	0.138 $\pm$ 0.040	41.91 $\pm$ 11.70	0.07 $\pm$ 0.028
	Proposed	132.48 $\pm$ 21.52	0.138 $\pm$ 0.021	19.84 $\pm$ 0.31	0.19 $\pm$ 0.101

Table 3. Cancer subtype classification metrics (%). Best results are in **bold**.

Input Modality	Accuracy	Precision	Recall	F1-score
H&E only (Baseline)	98.28 $\pm$ 2.20	98.61 $\pm$ 1.56	98.38 $\pm$ 1.99	98.42 $\pm$ 1.90
Virtual IHC (Proposed)	98.69 $\pm$ 0.70	98.83 $\pm$ 0.57	98.77 $\pm$ 0.59	98.78 $\pm$ 0.58
H&E + Virtual IHC (Proposed)	99.31 $\pm$ 0.56	99.36 $\pm$ 0.53	99.35 $\pm$ 0.55	99.35 $\pm$ 0.55

As shown in Table 3, training in the fused modalities yielded slight improvements in all metrics. In particular, the network trained solely on virtual IHC demonstrated significantly lower variance across folds compared to the H&E baseline. This variance reduction suggests that the generative process acts as a domain normalizer, mitigating acquisition noise and staining inconsistencies for more stable classification training.

4 Conclusion

We presented a virtual staining framework for unpaired H&E-to-PAS and IHC translations in renal pathology. Our quantitative findings highlight a fundamental Pareto trade-off: enforcing strict source-image morphology intrinsically restricts the diffusion model from perfectly matching the unconstrained target-stain color distribution. However, prioritizing structural fidelity over pure domain realism effectively prevents morphological hallucinations. Downstream experiments confirm that virtual IHC augmentation improves subcancer classification accuracy, demonstrating its potential to reduce costs and accelerate clinical workflows. Future work will explore weakly paired data, multi-site validation, and broader biomarker panels.

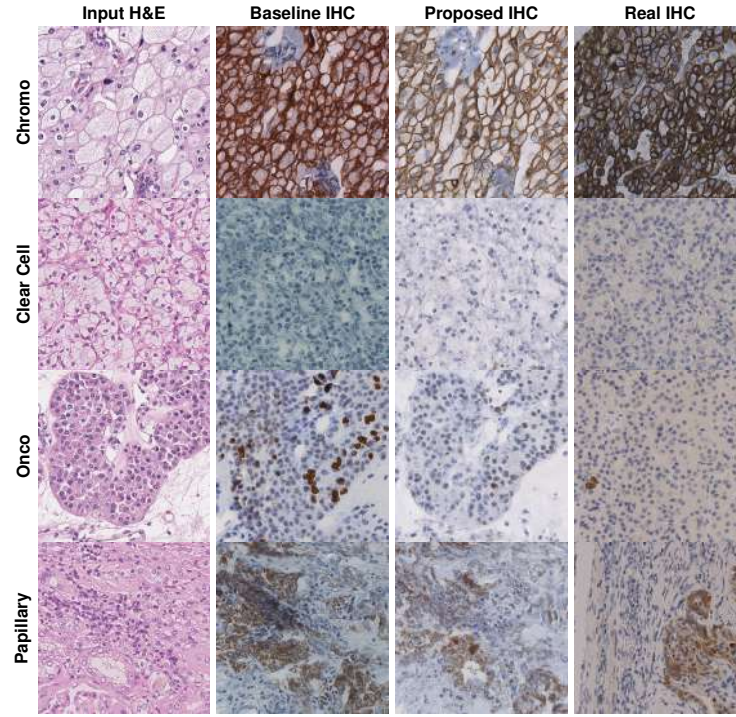


Fig. 3. H&E to IHC translation. The proposed pipeline preserves histological fidelity and prevents structural hallucinations (e.g., misaligned nuclei) seen in the baseline. Note: *Real IHC* patches are from different patients and serve solely as qualitative visual references, reflecting the unpaired nature of the data and the inherent inter-patient variability of IHC staining.

Acknowledgments. This work has been supported by the ANR Morpheus (263702) funding and the France 2030 investment plan managed by the Agence Nationale de la Recherche.

Disclosure of Interests. The authors have no competing interests.

References

1. Ahmed, A.A., Abedalthagafi, M.: Cancer diagnostics: the journey from histomorphology to molecular profiling. *Oncotarget* **7**(36), 58696 (2016). <https://doi.org/10.18632/oncotarget.11061>
2. Alwahaibi, N., Al Mamari, A.M., Al Aamri, A., Al Sulimani, Y., Al Balushi, A.: Assessing the diagnostic accuracy of routine hematoxylin and eosin, alcian blue/periodic acid-schiff, and giemsa stains in the detection of helicobacter pylori in gastric biopsies. *F1000Research* **14**, 1148 (2026). <https://doi.org/10.12688/f1000research.170290.2>

3. Bancroft, J.D., Gamble, M.: Theory and practice of histological techniques. Elsevier health sciences (2008)
4. Bińkowski, M., Sutherland, D.J., Arbel, M., Gretton, A.: Demystifying mmd gans. arXiv preprint arXiv:1801.01401 (2018). <https://doi.org/10.48550/arXiv.1801.01401>
5. Goldsborough, T., Philips, B., O’Callaghan, A., Inglis, F., Leplat, L., Filby, A., Bilen, H., Bankhead, P.: Instanseg: an embedding-based instance segmentation algorithm optimized for accurate, efficient and portable cell segmentation. arXiv preprint arXiv:2408.15954 (2024). <https://doi.org/10.48550/arXiv.2408.15954>
6. He, K., Sun, J., Tang, X.: Guided image filtering. IEEE transactions on pattern analysis and machine intelligence **35**(6), 1397–1409 (2013). <https://doi.org/10.1109/TPAMI.2012.213>
7. Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., Hochreiter, S.: Gans trained by a two time-scale update rule converge to a local nash equilibrium. Advances in neural information processing systems **30** (2017). <https://doi.org/10.48550/arXiv.1706.08500>
8. Li, Z., Su, T., Zhang, B., Han, W., Zhang, S., Sun, G., Cong, Y., Chen, X., Qi, J., Wang, Y., et al.: His-mmmdm: Multi-domain and multi-omics translation of histopathological images with diffusion models. Advanced Science **13**(19), e18066 (2026). <https://doi.org/10.1002/advs.202518066>
9. Magaki, S., Hojat, S.A., Wei, B., So, A., Yong, W.H.: An introduction to the performance of immunohistochemistry. In: Biobanking: methods and protocols, pp. 289–298. Springer (2018). https://doi.org/10.1007/978-1-4939-8935-5_25
10. Meng, C., He, Y., Song, Y., Song, J., Wu, J., Zhu, J.Y., Ermon, S.: Sdedit: Guided image synthesis and editing with stochastic differential equations. arXiv preprint arXiv:2108.01073 (2021). <https://doi.org/10.48550/arXiv.2108.01073>
11. Müller-Franzes, G., Niehues, J.M., Khader, F., Arasteh, S.T., Haarburger, C., Kuhl, C., Wang, T., Han, T., Nolte, T., Nebelung, S., et al.: A multimodal comparison of latent denoising diffusion probabilistic models and generative adversarial networks for medical image synthesis. Scientific reports **13**(1), 12098 (2023). <https://doi.org/10.1038/s41598-023-39278-0>
12. Reinhard, E., Adhikhmin, M., Gooch, B., Shirley, P.: Color transfer between images. IEEE Computer graphics and applications **21**(5), 34–41 (2001). <https://doi.org/10.1109/38.946629>
13. Rubner, Y., Tomasi, C., Guibas, L.J.: The earth mover’s distance as a metric for image retrieval. International journal of computer vision **40**(2), 99–121 (2000). <https://doi.org/10.1023/A:1026543900054>
14. Schmidt, U., Weigert, M., Broaddus, C., Myers, G.: Cell detection with star-convex polygons. In: International conference on medical image computing and computer-assisted intervention. pp. 265–273. Springer (2018). https://doi.org/10.1007/978-3-030-00934-2_30
15. Xiong, B., Peng, Y., Zhang, R., Chen, F., He, J., Qin, W.: Unpaired multi-domain histopathology virtual staining using dual path prompted inversion. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 39, pp. 8780–8787 (2025). <https://doi.org/10.1609/aaai.v39i8.32949>
16. Yellapragada, S., Graikos, A., Li, Z., Triaridis, K., Belagali, V., Kapse, S., Nandi, T.N., Madduri, R.K., Prasanna, P., Kurc, T., et al.: Pixcell: A generative foundation model for digital histopathology images. arXiv preprint arXiv:2506.05127 (2025). <https://doi.org/10.48550/arXiv.2506.05127>